

Practical 7

Process : process, shell process, process commands (ps and its options), utilities:
ps command

1. ps

```
student@student-virtual-machine:~$ ps
  PID TTY          TIME CMD
 3053 pts/0    00:00:00 bash
 4511 pts/0    00:00:00 ps
```

2. ps -l

```
student@student-virtual-machine:~$ ps -l
F S  UID        PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1000        3053   3035  0  80   0 -   2783 do_wai pts/0    00:00:00 bash
0 R  1000        4512   3053  0  80   0 -   3167 -      pts/0    00:00:00 ps
```

3. ps -e

```
student@student-virtual-machine:~$ ps -e
  PID TTY          TIME CMD
   1 ?           00:00:01 systemd
   2 ?           00:00:00 kthreadd
   3 ?           00:00:00 pool_workqueue_release
   4 ?           00:00:00 kworker/R-rcu_g
   5 ?           00:00:00 kworker/R-rcu_p
   6 ?           00:00:00 kworker/R-slub_
   7 ?           00:00:00 kworker/R-netns
   9 ?           00:00:00 kworker/0:0H-kblockd
  12 ?           00:00:00 kworker/R-mm_pe
  13 ?           00:00:00 rcu_tasks_kthread
  14 ?           00:00:00 rcu_tasks_rude_kthread
  15 ?           00:00:00 rcu_tasks_trace_kthread
  16 ?           00:00:00 ksoftirqd/0
  17 ?           00:00:00 rcu_preempt
  18 ?           00:00:00 migration/0
  19 ?           00:00:00 idle_inject/0
  20 ?           00:00:00 cpuhp/0
  21 ?           00:00:00 cpuhp/1
  22 ?           00:00:00 idle_inject/1
  23 ?           00:00:00 migration/1
  24 ?           00:00:00 ksoftirqd/1
  29 ?           00:00:00 kdevtmpfs
  30 ?           00:00:00 kworker/R-inet_
```

4. ps -ef

```
student@student-virtual-machine:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1        0  0  17:33 ?        00:00:01 /sbin/init auto noprompt splash
root           2        0  0  17:33 ?        00:00:00 [kthreadd]
root           3        2  0  17:33 ?        00:00:00 [pool_workqueue_release]
root           4        2  0  17:33 ?        00:00:00 [kworker/R-rcu_g]
root           5        2  0  17:33 ?        00:00:00 [kworker/R-rcu_p]
root           6        2  0  17:33 ?        00:00:00 [kworker/R-slub_]
root           7        2  0  17:33 ?        00:00:00 [kworker/R-netns]
root           9        2  0  17:33 ?        00:00:00 [kworker/0:0H-kblockd]
root          12        2  0  17:33 ?        00:00:00 [kworker/R-mm_pe]
root          13        2  0  17:33 ?        00:00:00 [rcu_tasks_kthread]
root          14        2  0  17:33 ?        00:00:00 [rcu_tasks_rude_kthread]
root          15        2  0  17:33 ?        00:00:00 [rcu_tasks_trace_kthread]
root          16        2  0  17:33 ?        00:00:00 [ksoftirqd/0]
root          17        2  0  17:33 ?        00:00:00 [rcu_preempt]
root          18        2  0  17:33 ?        00:00:00 [migration/0]
root          19        2  0  17:33 ?        00:00:00 [idle_inject/0]
root          20        2  0  17:33 ?        00:00:00 [cpuhp/0]
root          21        2  0  17:33 ?        00:00:00 [cpuhp/1]
root          22        2  0  17:33 ?        00:00:00 [idle_inject/1]
root          23        2  0  17:33 ?        00:00:00 [migration/1]
root          24        2  0  17:33 ?        00:00:00 [ksoftirqd/1]
root          29        2  0  17:33 ?        00:00:00 [kdevtmpfs]
root          30        2  0  17:33 ?        00:00:00 [kworker/R-inet_]
```

5. ps -F

```
student@student-virtual-machine:~$ ps -F
UID          PID    PPID  C  SZ  RSS  PSR  STIME TTY          TIME CMD
student    3053    3035  0 2783 5284   1 17:36 pts/0    00:00:00 bash
student    4558    3053  0 3167 3548   0 19:24 pts/0    00:00:00 ps -F
student@student-virtual-machine:~$ ps -ef
UID          PID    PPID  C  SZ  RSS  PSR  STIME TTY          TIME CMD
root           1        0  0 44032 13380   1 17:33 ?        00:00:01 /sbin/init auto noprompt splash
root           2        0  0      0      0   0  0 17:33 ?        00:00:00 [kthreadd]
root           3        2  0      0      0   0  0 17:33 ?        00:00:00 [pool_workqueue_release]
root           4        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/R-rcu_g]
root           5        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/R-rcu_p]
root           6        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/R-slub_]
root           7        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/R-netns]
root           9        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/0:0H-kblockd]
root          12        2  0      0      0   0  0 17:33 ?        00:00:00 [kworker/R-mm_pe]
root          13        2  0      0      0   0  0 17:33 ?        00:00:00 [rcu_tasks_kthread]
root          14        2  0      0      0   0  0 17:33 ?        00:00:00 [rcu_tasks_rude_kthread]
```